

Please write clearly in block capitals.

Centre number

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Candidate number

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Surname

Forename(s)

Candidate signature

INTERNATIONAL A-LEVEL CHEMISTRY (9620)

Unit 3: Inorganic 2 and Physical 2

Tuesday 4 June 2019

07:00 GMT

Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:

- the Periodic Table/Data Sheet, provided as an insert
- a ruler with millimetre measurements
- a scientific calculator, which you are expected to use where appropriate.

Instructions

- Use black ink or black ball-point pen. Use pencil only for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- All working must be shown.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 80.

For Examiner's Use	
Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
TOTAL	



Answer **all** questions in the spaces provided.

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0 1

This question is about two metal iodides.

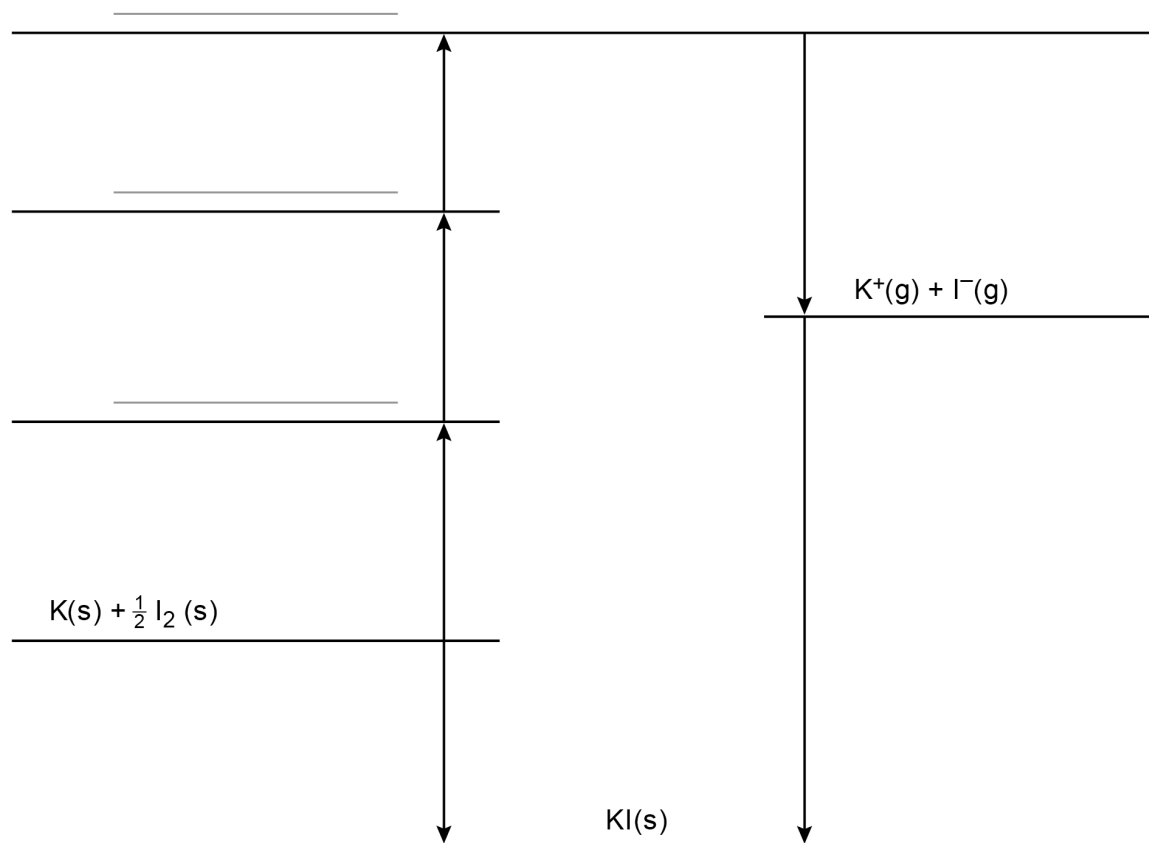
Table 1 shows some thermodynamic data.

Table 1

Enthalpy change	$\Delta H^\ominus / \text{kJ mol}^{-1}$
Atomisation of potassium	+90
First ionisation energy of potassium	+418
Atomisation of iodine	+107
Electron affinity of iodine	−314
Lattice formation of potassium iodide	−629

Figure 1 shows an incomplete Born–Haber cycle for the formation of potassium iodide.

Figure 1



0 1 . 1

State how **Figure 1** shows that the cycle is for the enthalpy of lattice formation, **not** the enthalpy of lattice dissociation.

[1 mark]

0 1 . 2

Complete **Figure 1** by writing the missing symbols for the species on the appropriate energy levels.

[3 marks]

0 1 . 3

Use **Figure 1** and the data in **Table 1** to calculate a value for the enthalpy of formation of potassium iodide.

[2 marks]

Enthalpy of formation _____ kJ mol^{-1}

Question 1 continues on the next page

Turn over ►



0 1 . 4

The value obtained in Question **01.3** is based on experimental data and using a Born–Haber cycle.

Enthalpies of lattice formation can also be determined by calculation using a perfect ionic model.

Table 2 shows the enthalpies of lattice formation for potassium iodide and silver iodide.

Table 2

Enthalpy change	$\Delta H^\ominus / \text{kJ mol}^{-1}$	
	Born–Haber cycle	Perfect ionic model
Lattice formation of potassium iodide	–629	–632
Lattice formation of silver iodide	–876	–736

State what can be deduced about the bonding in each compound from the data in **Table 2**

[2 marks]

Potassium iodide _____

Silver iodide _____



0 1 . 5 **Table 3** shows some thermodynamic data.

Table 3

Enthalpy change	$\Delta H^\ominus / \text{kJ mol}^{-1}$
Hydration of K^+ (g)	−322
Hydration of I^- (g)	−293
Lattice formation of KI (s)	−629

Use the data in **Table 3** to calculate the enthalpy of solution of potassium iodide.

[2 marks]

Enthalpy of solution _____ kJ mol^{-1}

10

Turn over for the next question

Turn over ►



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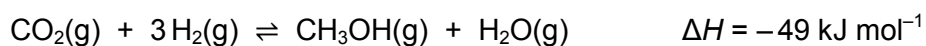
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ANSWER IN THE SPACES PROVIDED**



0 2

Methanol is made in industry as shown.



0 2 . 1

State why a higher equilibrium yield of methanol is obtained at a lower temperature.

[1 mark]

0 2 . 2

Predict whether a high or low pressure increases the equilibrium yield of methanol.
Give a reason for your prediction.

[2 marks]

Prediction

Reason

0 2 . 3

A catalyst is used in this industrial reaction.

What happens if a catalyst is **not** used?Tick (✓) **one** box.

[1 mark]

The position of equilibrium will shift to the right.

☐

The value of enthalpy change will stay the same.

☐

The yield of methanol at equilibrium will decrease.

☐

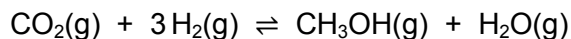
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0 2 . 4

A mixture of 15.0 mol of carbon dioxide and 45.0 mol of hydrogen is heated in a container. At equilibrium, the mixture contains 6.3 mol of steam.

Calculate the amount, in moles, of each of the other substances in the equilibrium mixture.



[3 marks]

Amount of $\text{CH}_3\text{OH}(\text{g})$ _____ mol

Amount of $\text{CO}_2(\text{g})$ _____ mol

Amount of $\text{H}_2(\text{g})$ _____ mol

0 2 . 5

Table 4 shows the amount, in moles, of each substance in the equilibrium mixture at a different temperature and at a total pressure of 2.50×10^6 Pa

Table 4

Substance	$\text{CO}_2(\text{g})$	$\text{H}_2(\text{g})$	$\text{CH}_3\text{OH}(\text{g})$	$\text{H}_2\text{O}(\text{g})$
Amount / mol	4.9	7.6	19.4	13.7

Calculate the partial pressure of methanol in this equilibrium mixture.

[2 marks]

Partial pressure of methanol _____ Pa



0 2 . 6

Write an expression for the equilibrium constant (K_p) for this reaction.

[1 mark]

 K_p

0 2 . 7

Table 5 shows the partial pressures, in Pa, of each substance in another equilibrium mixture.**Table 5**

Substance	CO ₂ (g)	H ₂ (g)	CH ₃ OH(g)	H ₂ O(g)
Partial pressure / Pa	1.68×10^6	2.64×10^6	8.34×10^6	1.76×10^6

Use the data in **Table 5** to calculate the value of K_p
Deduce the units of K_p

[2 marks]

 K_p _____ units _____

12

Turn over for the next question

Turn over ►



0 3

Carbon dioxide changes directly from a solid to a gas at atmospheric pressure.

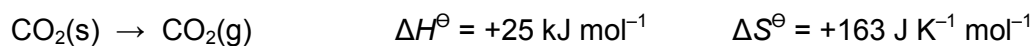
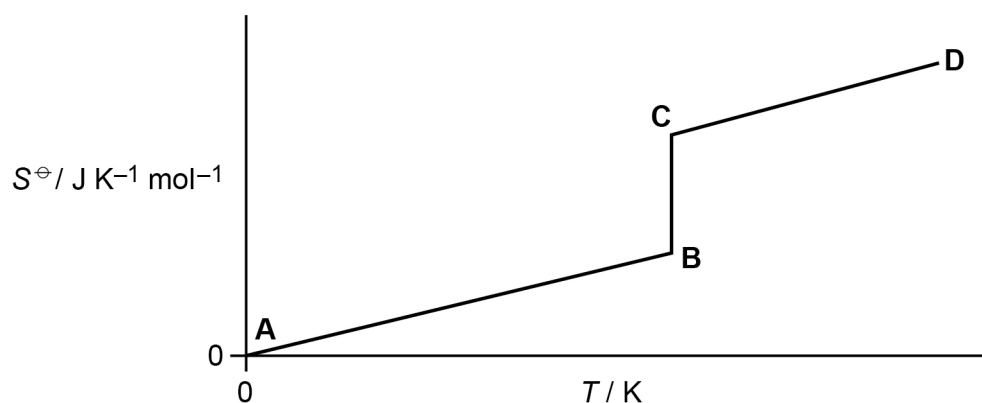


Figure 2 shows how the entropy of carbon dioxide changes with temperature.

Figure 2



0 3 . 1

State why the value of $\Delta H^\ominus$ is positive for the change of state of carbon dioxide from solid to gas.

[1 mark]

0 3 . 2

State why carbon dioxide has zero entropy at a temperature of 0 K

[1 mark]



0 3 . 3

Describe how the movement of the molecules changes during the three stages shown on **Figure 2**

[3 marks]

A to B _____

B to C _____

C to D _____

0 3 . 4

Calculate the temperature, in K, at which the Gibbs free-energy change, ΔG , has a value of zero for the change of state of carbon dioxide from solid to gas.

[2 marks]

Temperature _____ K

7

Turn over for the next question

Turn over ►



0 4

This question is about determining a rate equation and an activation energy from experimental data.

The reaction between compounds **E** and **F** was investigated.

The data in **Table 6** were obtained in three experiments at a constant temperature.

Table 6

Experiment	Initial [E] / mol dm ⁻³	Initial [F] / mol dm ⁻³	Initial rate / mol dm ⁻³ min ⁻¹
1	1.3×10^{-2}	2.8×10^{-2}	3.2×10^{-3}
2	3.9×10^{-2}	2.8×10^{-2}	9.6×10^{-3}
3	3.9×10^{-2}	1.4×10^{-2}	2.4×10^{-3}

0 4 . 1

Use the data in **Table 6** to deduce the order of reaction with respect to **E** and the order of reaction with respect to **F**.

Determine the rate equation for this reaction.
Show your working.

[3 marks]

Order with respect to **E** _____

Order with respect to **F** _____

Rate equation _____



0 4 . 2

Use data from **Table 6** to calculate a value for the rate constant (k), including units, for the reaction between **E** and **F**.

(If you were unable to determine a rate equation in Question **04.1**, you should assume that the equation is $rate = k[E]^2[F]^2$. This is not the correct equation.)

[2 marks]

k _____ units _____

Compound **G** decomposes on heating in a first order reaction.

The decomposition of **G** in a given volume of solution is investigated.

0 4 . 3

State why the rate of reaction doubles when the concentration of **G** doubles at a constant temperature.

[1 mark]

0 4 . 4

Explain why the rate of reaction more than doubles when the temperature is increased by 20 °C at a constant concentration.

[2 marks]

Turn over ►

0 4 . 5

The Arrhenius equation can be used to determine the activation energy (E_a) for the decomposition of **G**.

$$\ln k = \frac{-E_a}{RT} + \ln A$$

The rate constant for the decomposition of **G**, $k = 0.273 \text{ min}^{-1}$ at 400 K

The Arrhenius constant, $A = 3.27 \times 10^8 \text{ min}^{-1}$

The gas constant, $R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$

Use these data to determine a value for E_a (in kJ mol^{-1}) for the decomposition at 400 K

[3 marks]

E_a _____ kJ mol^{-1}



Turn over for the next question

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ANSWER IN THE SPACES PROVIDED**

Turn over ►



0 5

At high temperatures, aluminium chloride exists as gaseous AlCl_3 molecules.
On cooling, two molecules of AlCl_3 combine to form one molecule of Al_2Cl_6

0 5 . 1

Name the shape formed by the atoms in an AlCl_3 molecule.

State the bond angle in AlCl_3

[2 marks]

Name of shape _____

Bond angle _____

0 5 . 2

Co-ordinate bonds form when two AlCl_3 molecules combine together.

State how one of these bonds forms.

[1 mark]

0 5 . 3

Draw the structure of an Al_2Cl_6 molecule.

Show each covalent bond as a line (—) and each co-ordinate bond as an arrow ($\rightarrow$).

[2 marks]

0	5	.	4
---	---	---	---

Some anhydrous aluminium chloride is added to an excess of water.

Describe what is observed.

Write an equation for the reaction that occurs.

Suggest a possible value for the pH of the solution.

[3 marks]

Observation _____

Equation

pH of solution _____

0	5	.	5
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A student added an excess of sodium carbonate solution to aluminium chloride solution.

State **two** observations the student would make.

Give an equation for the reaction that occurs.

[3 marks]

Observation 1 _____

Observation 2 _____

Equation

11

Turn over for the next question

Turn over ►



0 6**Table 7** shows the melting points of some Period 3 chlorides.**Table 7**

	MgCl₂	SiCl₄
Melting point / K	987	204

Use your knowledge of structure and bonding to answer Questions **06.1** and **06.2****0 6 . 1**

Explain why magnesium chloride has a high melting point.

[2 marks]

0 6 . 2

Explain why silicon(IV) chloride has a low melting point.

[2 marks]

0 6 . 3

Both silicon(IV) chloride and phosphorus(V) chloride react with water to form a solution containing hydrochloric acid.

Write an equation for each reaction.

[2 marks]

Equation for silicon(IV) chloride reaction

Equation for phosphorus(V) chloride reaction

6

0 7

This question is about properties of transition metal compounds.

0 7 . 1

Transition metal complexes can change colour when the oxidation state changes.

Give **two** other factors that can cause a change in the colour of a transition metal complex.

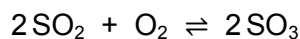
[2 marks]

1 _____

2 _____

0 7 . 2

The equation for the reaction that occurs in the Contact process is shown.



Write **two** equations to show how vanadium(V) oxide acts as a catalyst in this reaction.

Give the change in oxidation state of vanadium in equation 2.

[2 marks]

Equation 1

Equation 2

Change in oxidation state _____

Question 7 continues on the next page

Turn over ►

0 7 . 3 Vanadium(V) oxide is a heterogeneous catalyst.

Give the meaning of the term heterogeneous.

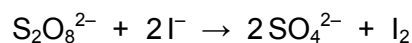
Suggest how the effect of a heterogeneous catalyst can be increased without using a higher temperature.

[2 marks]

Meaning of heterogeneous _____

How the effect can be increased _____

0 7 . 4 Iodide ions react slowly with peroxodisulfate ions ($\text{S}_2\text{O}_8^{2-}$) in aqueous solution as shown.



This reaction can be catalysed by adding an aqueous solution containing Fe^{2+} ions.

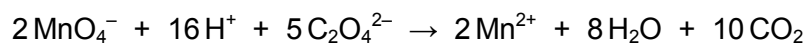
Explain, with the aid of equations, how Fe^{2+} ions catalyse the reaction between I^- ions and $\text{S}_2\text{O}_8^{2-}$ ions.

[4 marks]



0 7 . 5

The overall equation for the oxidation of ethanedioate ions by manganate(VII) ions is shown.

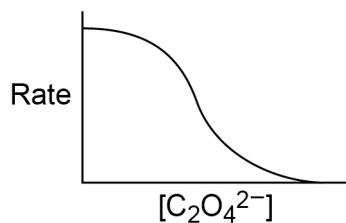
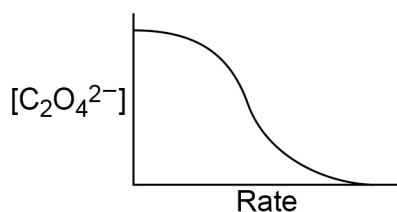
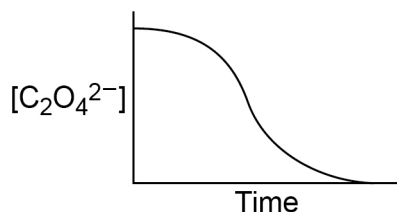
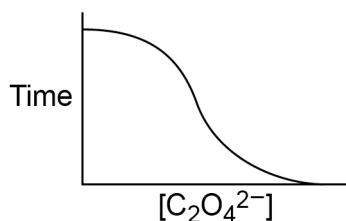


This reaction is an example of autocatalysis.

Which graph represents the change that occurs during this reaction?

Tick (✓) **one** box.

[1 mark]


☐

☐

☐

☐

0 7 . 6

A student adds an excess of a solution containing ethanedioate ions to a solution containing manganate(VII) ions.

State **one** observation that the student would make.

[1 mark]

12

Turn over ►



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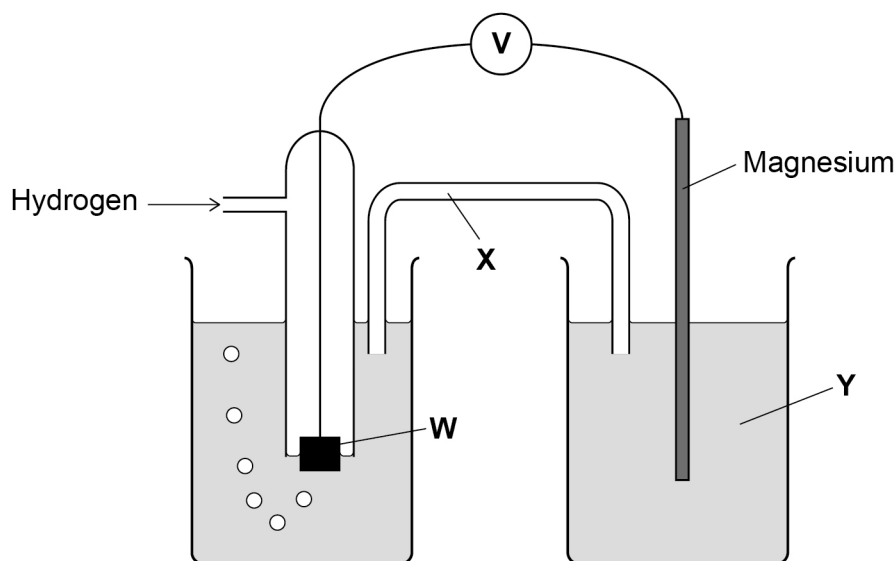
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0 8

Figure 3 shows a cell used to measure the standard electrode potential of the Mg^{2+}/Mg half-cell.

Figure 3



0 8 . 1

Identify a suitable substance that could be used for each of **W**, **X** and **Y**.

[3 marks]

W _____

X _____

Y _____

0 8 . 2

Identify **two** conditions needed in this cell for the measurement to be a **standard** electrode potential.

[2 marks]

1 _____

2 _____

Question 8 continues on the next page

Turn over ►



Table 8 shows some standard electrode potential data.

Table 8

Electrode half-equation	$E^\ominus / \text{V}$
$\text{VO}_2^+(\text{aq}) + 2\text{H}^+(\text{aq}) + \text{e}^- \rightarrow \text{VO}^{2+}(\text{aq}) + \text{H}_2\text{O}(\text{l})$	+1.00
$\text{Fe}^{3+}(\text{aq}) + \text{e}^- \rightarrow \text{Fe}^{2+}(\text{aq})$	+0.77
$\text{VO}^{2+}(\text{aq}) + 2\text{H}^+(\text{aq}) + \text{e}^- \rightarrow \text{V}^{3+}(\text{aq}) + \text{H}_2\text{O}(\text{l})$	+0.34
$\text{Cr}^{3+}(\text{aq}) + \text{e}^- \rightarrow \text{Cr}^{2+}(\text{aq})$	-0.41
$\text{Fe}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Fe}(\text{s})$	-0.44
$\text{Zn}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Zn}(\text{s})$	-0.76
$\text{Cr}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cr}(\text{s})$	-0.91

0 8 . 3 Identify the strongest reducing agent in **Table 8**

[1 mark]

0 8 . 4 Identify the species in **Table 8** that will oxidise $\text{V}^{3+}(\text{aq})$ to $\text{VO}^{2+}(\text{aq})$ but will **not** oxidise $\text{VO}^{2+}(\text{aq})$ to $\text{VO}_2^+(\text{aq})$.

[1 mark]

0 8 . 5 An excess of solid zinc is added to a solution containing iron(III) ions.

Use the data in **Table 8** to explain why a reaction occurs.
Identify the products.

[2 marks]

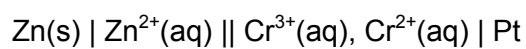
Explanation _____

Products _____



0	8	.	6
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The conventional representation of a standard cell is shown.



Calculate the EMF of this cell.

Deduce an equation for the overall cell reaction.

[2 marks]

EMF _____

Equation

11

END OF QUESTIONS



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